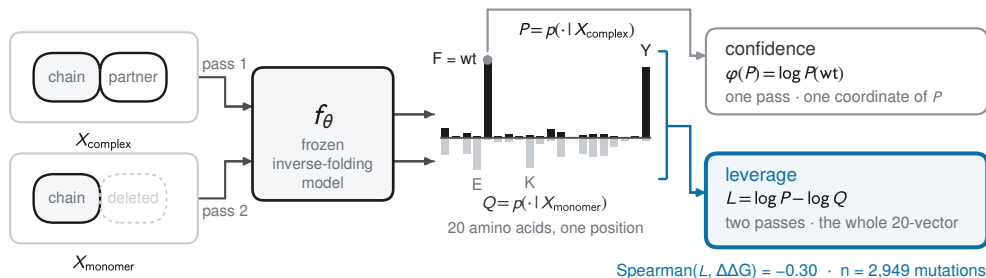


a confidence reads one coordinate of P; leverage reads the whole $P \rightarrow Q$ move

example position: 3SZK chain C F44 (a hotspot) · bars are the real 20-simplex



diffusion $\varepsilon_\theta(x | c)$ — $\varepsilon_\theta(x | \emptyset)$ classifier-free guidance
this work $\log p(\cdot | X_{\text{complex}})$ — $\log p(\cdot | X_{\text{monomer}})$ is this same difference

b confidence does not pin leverage down

13,401 positions · 343 complexes · 327 hotspots

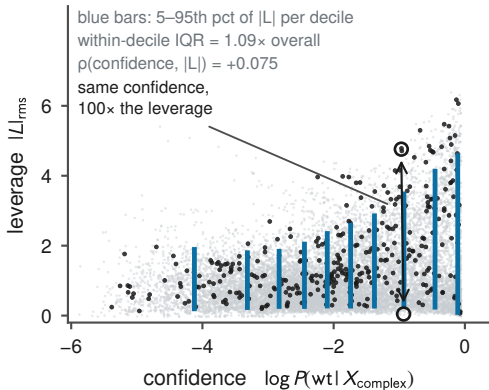
blue bars: 5–95th pct of $|L|$ per decile

within-decile IQR = 1.09× overall

$\rho(\text{confidence}, |L|) = +0.075$

same confidence,

100× the leverage

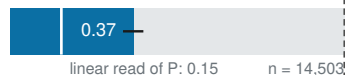


c

and neither does all of P

GBM / random forest, complex-held-out

ProteinMPNN 63% irreducible



ESM-IF1 64% irreducible



L unrelated to P

P would determine L

R^2 of $|L|_{\text{rms}}$ from all 20 coordinates of P

0.0 0.5 1.0